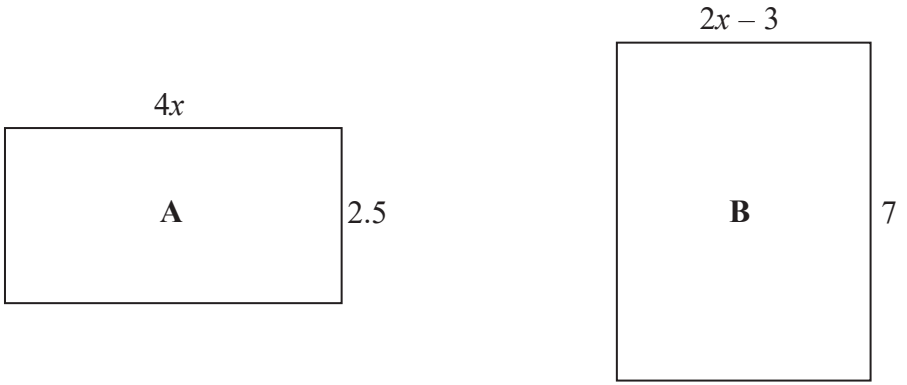
21 Here are two rectangles.



All measurements are in centimetres.

The area of rectangle **A** is equal to the area of rectangle **B**.

Work out the perimeter of rectangle **B**.

1 mark
Form area expressions

×	$2x$	-3
7	$14x$	-21

Area A = 2.5 × 4x = 10x.

1 mark
Equate areas

10x = 14x – 21

(-10x) (-10x)

0 = 4x – 21

(+21) (+21)

21 = 4x

(÷4) (÷4)

1 mark
Solve equation

x = 5.25; width B = 2x – 3 = 7.5cm.

1 mark
x=5.25

Perimeter B = 2(7 + 7.5) = 29cm.

1 mark
29cm perimeter

.....cm

(Total for Question 21 is 5 marks)